



MA 255: Number Theory

Course Code	MA 255
Course Name	Number Theory
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	B.Tech. in Mathematics and Computing
Distribution	Discipline elective
Mutual Exclusion	MA 552
Approval	59th BoA (Apr 2025), in the B.Tech Mathematics and Computing curriculum; 61st BoA (Jan 2026)

Preamble

Elementary Number Theory is a fundamental area of mathematics that develops deep insight into the structure and properties of integers. Concepts originating from number theory form the theoretical backbone of several areas in modern mathematics and computing, including algorithms, cryptography, coding theory, and computational complexity. The proposed course on Elementary Number Theory is designed to introduce students to core topics such as divisibility, prime numbers, greatest common divisors, congruences, modular arithmetic, Diophantine equations, and arithmetic functions. Emphasis is placed on rigorous reasoning, proof techniques, and the development of algorithmic methods for solving number-theoretic problems.

Course Modules

- **Unit 1:** Divisibility, The greatest common divisor, The fundamental theorem of integers, Basic properties of congruences, Some special congruences, Polynomial congruences and their solutions, Primitive roots (16 Hours)
- **Unit 2:** Quadratic residues and their basic properties, The Gauss Lemma, The law of quadratic reciprocity, Applications to Diophantine equations, Multiplicative arithmetic functions, The Mobius inversion formula, Perfect and Amicable numbers. (16 Hours)
- **Unit 3:** Integer representation and algorithms, Hashing functions, Pseudorandom numbers, Check digits, Introduction to classic cryptography. (10 Hours)

Text Books

1. William W. Adams, Larry J. Goldstein, *Introduction to Number Theory*, Princeton-Hall Inc., 1976
2. Kenneth H. Rosen, *Discrete Mathematics and its applications*, 8th edition, 2021.